ABSTRACT

Sign language is an essential means of communication for people with hearing and speech impairments. However, communication between sign language users and non-sign language users often becomes difficult due to the lack of widespread understanding of sign language. A **Real-Time Sign Language Recognition System** aims to bridge this communication gap by automatically detecting and translating hand gestures into readable text or audible speech using advanced computer vision and machine learning techniques. This system captures hand movements and gestures through a camera and processes them instantly using image processing algorithms and deep learning models.

The proposed system utilizes real-time video input to recognize sign language gestures accurately and efficiently. Initially, the camera captures hand gestures, and the system applies preprocessing techniques such as background removal, hand segmentation, and image normalization to isolate the hand region. After preprocessing, feature extraction methods are used to identify key points, shapes, and movement patterns of the hand. These extracted features are then fed into a trained machine learning or deep learning model, such as Convolutional Neural Networks (CNNs), to classify the gestures into corresponding letters, words or phrases. The recognized gesture is then converted into text and optionally into speech output, allowing seamless communication with people who do not understand sign language.

The real-time nature of the system ensures that gestures are recognized instantly, making the interaction natural and efficient. Such systems can be implemented using affordable hardware like webcams or smartphone cameras, making the technology accessible and scalable. Additionally, the integration of modern artificial intelligence techniques improves recognition accuracy and reduces errors caused by variations in lighting, hand orientation and user differences.

This system has significant potential in improving accessibility and inclusivity for the deaf and hard-of-hearing community. It can be used in educational institutions, public service centres, healthcare facilities and everyday communication scenarios. By enabling automatic translation of sign language into text or speech, the **Real-Time Sign Language Recognition System** promotes better social integration and enhances communication between sign language users and the general population.